Temporal Prototype Alignment for Frozen-Feature Dataset Distillation

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Abstract

Dataset distillation condenses large-scale training sets into compact synthetic data. Recent frozen-feature distillation methods, such as Linear Gradient Matching, make ImageNet-scale synthesis feasible by matching gradients in pretrained representation spaces, but their reliance on exhaustive spatial view ensembles incurs memory and I/O costs that grow linearly with the number of augmentations. We propose Temporal Prototype Alignment (TPA), a temporal gradient estimation framework for frozen-feature dataset distillation. TPA replaces spatial Monte Carlo gradient averaging with a temporal prototype estimator that accumulates stable semantic signals across optimization steps. An adaptive momentum gate modulates the estimator according to local signal reliability, reducing gradient variance while keeping memory complexity independent of the number of spatial views. To improve cross-architecture transfer, we introduce a consensus alignment objective that distills shared structure from multiple pretrained teachers while mitigating model-specific artifacts. Across ImageNet-scale and fine-grained benchmarks, TPA substantially reduces memory overhead, enables distillation on consumer-grade GPUs, and improves transfer to heterogeneous CNN and Vision Transformer backbones. These results suggest that online temporal estimation provides an efficient alternative to spatial ensembling for frozen-feature dataset distillation.
Project page: https://ruiding1.github.io/TPA-project/

1 Introduction

Dataset distillation synthesizes a compact labeled dataset such that a model trained on the distilled data achieves performance comparable to training on the full dataset [1]. Traditional research has primarily focused on the training-from-scratch paradigm, in which synthetic data is optimized to supervise randomly initialized networks [2–6]. Meanwhile, modern vision pipelines increasingly favor linear probing over fixed, pretrained representations. This paradigm shift motivates dataset distillation methods that operate directly in frozen feature space and scale to large backbones.

A recent breakthrough, Linear Gradient Matching (LGM) [7] enabled ImageNet-scale distillation by transitioning from training-from-scratch to gradient alignment within frozen feature spaces. Despite its impressive performance at one image per class (IPC=1), LGM relies on

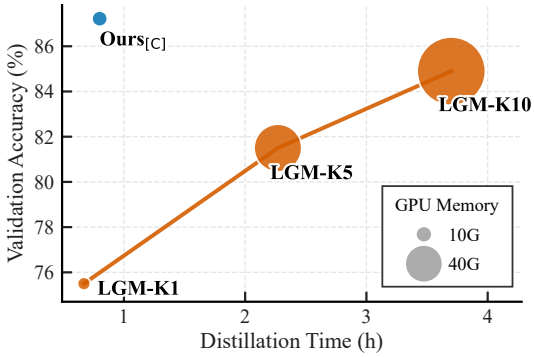
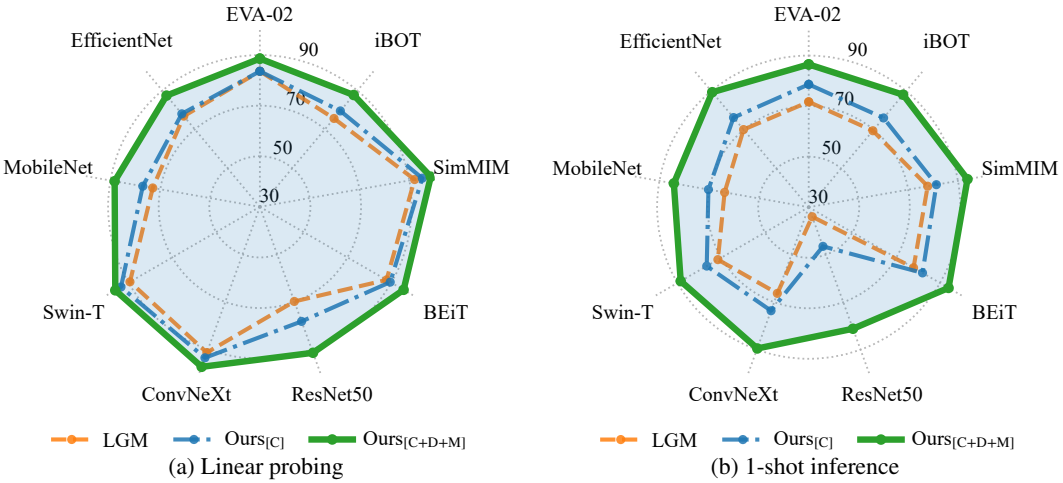


Figure 1: **Efficiency and Performance Comparison (IPC=1).** Our method achieves superior accuracy compared to LGM with substantially reduced training costs. Specifically, Ours[C] matches LGM with a 10× memory reduction. This efficiency enables Ours[C+D+M] ([C]: CLIP, [D]: DINOv2, [M]: MoCo-v3) to leverage cross-model consensus, achieving new SOTA results.

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(a) Linear probing (b) 1-shot inference

Figure 2: **Cross-architecture generalization and black-box transferability.** Ours$_{[C+D+M]}$ consistently improves transfer across 9 unseen heterogeneous target models under both linear probing and training-free 1-shot inference.

differentiable augmentation ensembles, typically using K=10 views per step. These multi-view augmentations stabilize gradient estimates and enforce cross-view consistency, thereby preventing view-specific shortcuts. However, this strategy incurs a linear surge in memory and I/O overhead, reaching 81 GB on ImageNet-100. Scaling to ImageNet-1K further requires enterprise-grade resources, such as 4×H200 GPUs for over 12 hours. More critically, this memory bottleneck restricts existing methods to a single-teacher setting, thereby limiting their ability to generalize across architectures. Because any individual model inevitably encodes architecture-specific inductive biases, synthetic data distilled from a single teacher remain constrained by these biases, irrespective of augmentation intensity. This limitation fundamentally challenges the scalability and transferability of current frozen-feature distillation paradigms.

In this paper, we challenge the necessity of computationally expensive spatial ensembling strategies. Our key observation is that, although gradients computed from single augmented views are inherently stochastic and view-dependent, their temporal trajectories preserve stable semantic signals under diverse augmentations. By shifting from spatial ensembling to temporal integration, we reconstruct robust distillation prototypes from historical gradient information, thereby eliminating the substantial computational and memory overhead associated with multi-view processing. Moreover, the resulting efficiency enables distillation across heterogeneous models, aligning with the Platonic Representation Hypothesis [8], which posits that diverse large-scale models converge toward shared representations.

Based on this insight, we introduce Adaptive Temporal Momentum, a gating mechanism that modulates gradient accumulation based on the local signal-to-noise ratio. This mechanism dynamically suppresses high-variance gradient outliers while rapidly adapting to consistent semantic signals, thus distilling stable prototypes from stochastic updates. To ensure that distilled data encode model-invariant knowledge, we propose Contrastive Consensus Alignment. Rather than matching gradients within a single architecture, we cast dataset distillation as a consensus-seeking process across heterogeneous backbones using a variational mutual information objective. This formulation ensures the synthesized images encapsulate universal patterns rather than model-specific biases.

The results are striking. As shown in Figure 1, our method distilled from CLIP achieves superior accuracy on ImageNet-100 using only 7 GB of GPU memory on an RTX 4090, compared to 81 GB required by LGM. This achievement represents a 10× memory reduction and 4× acceleration. Figures 2a and 2b confirm the consistent superiority of the proposed consensus framework over existing baselines in both linear probing and 1-shot inference across nine heterogeneous architectures, spanning both CNNs and ViTs. The main contributions are as follows:

- We propose a paradigm shift from spatial ensembling to temporal integration via a gating mechanism that reconstructs stable distillation prototypes from single-view gradients, substantially reducing memory and computational overhead.
- We propose contrastive consensus alignment to distill architecture-agnostic consensus knowledge from heterogeneous foundation models, significantly enhancing the cross-model transferability of the distilled images.

- Our method achieves new SOTA results on ImageNet-scale benchmarks under both linear probing and 1-shot inference, achieving a $4\times$ speedup and a $10\times$ memory reduction on a single consumer RTX 4090.

2 Related Work

Dataset Distillation. Dataset Distillation emerges as a promising solution by condensing large datasets into compact synthetic sets that retain the training efficacy [9]. Early work formulated distillation as a bi-level optimization problem, where the synthetic set is directly optimized by backpropagating through multiple inner-loop training steps [10–16]. Subsequent methods replaced long unrolled training with more tractable objectives, such as *gradient matching* [3, 17–20] for aligning update directions, and *trajectory matching* [21–23] for matching multi-step optimization trajectories. But, these approaches often struggle to scale due to the memory and instability inherent to bi-level optimization. Recent efforts have pushed condensation toward ImageNet scale, such as exact gradient computation for ImageNet-1K [24–27] and multi-stage large-scale condensation pipelines [28, 29]. Crucially, most existing literature assumes *training from scratch* on randomly initialized networks [30–33]. In contrast, LGM pioneers ImageNet-scale distillation within the frozen feature space. However, LGM relies on augmentation ensembles to stabilize gradients and enforce multi-view consistency, causing linear growth in GPU memory and I/O.

Our work shows that such expensive concurrent ensembles are unnecessary. Instead, we exploit *temporal integration* over stochastic single-view gradients to reconstruct stable distillation prototypes, while further promoting architecture-agnostic transfer via cross-backbone consensus

Vision Foundation Models. Self-supervised learning has become the dominant paradigm for learning transferable visual representations from unlabeled data. Major families include contrastive learning [34–37], self-distillation approaches such as DINO [38], and masked reconstruction methods exemplified by MAE [39–41]. In parallel, self-supervised learning has been successfully scaled to Vision Transformers, e.g., MoCo-v3 [42], while large-scale pre-training methods such as DINOv2 yield general-purpose visual features without supervision [43]. Additionally, vision-language models like CLIP further enrich representations through natural-language supervision, demonstrating strong transfer and zero-shot capabilities [44–46]. Under linear evaluation protocols, these diverse pre-training paradigms often produce similarly useful representations. This aligns with the *Platonic Representation Hypothesis* [8], which posits that as models and data scale, learned representations tend to converge toward a shared statistical structure across architectures and even modalities [47].

This observation motivates our design: data distilled from a single frozen backbone may inherit model-specific biases, whereas enforcing alignment across heterogeneous foundation models can distill model-invariant semantics. Thus, we formulate distillation as a cross-backbone consensus process, achieving generalization beyond any single backbone.

3 Method

3.1 Problem Formulation and Preliminaries

Problem Setup. Given a full real dataset $\mathcal{D} = \{(x^i_{\text{real}}, y^i_{\text{real}})\}_{i=1}^{|\mathcal{D}|}$, dataset distillation aims to synthesize a much smaller labeled dataset $\mathcal{S} = \{(x^i_{\text{syn}}, y^i_{\text{syn}})\}_{i=1}^{|\mathcal{S}|}$, where $|\mathcal{S}| \ll |\mathcal{D}|$, while preserving the essential knowledge of $\mathcal{D}$. Under the dominance of pre-trained vision foundation models, we follow the standard linear probing protocol: the feature extractor ϕ remains frozen to preserve the pre-trained representations, and the training process focuses on learning a linear classifier $W \in \mathbb{R}^{C\times F}$, where C denotes the number of classes and F is the feature dimension.

Revisiting Linear Gradient Matching. To obtain the discriminative information of $\mathcal{D}$ into $\mathcal{S}$, LGM aligns the gradients of the linear classifier derived from both datasets. At each distillation iteration, a linear classifier is randomly sampled $W \sim \mathcal{N}(0, 1)$, and the objective is formulated to minimize the cosine distance between the gradients:

$$\mathcal{L}_{\text{meta}} = 1 - \cos\left(\text{vec}(\nabla_W \ell_{\text{real}}), \text{vec}(\nabla_W \ell_{\text{syn}})\right), \tag{1}$$

where ℓ_{real} and ℓ_{syn} denote the cross-entropy losses evaluated on the real and synthetic datasets, respectively. To obtain a reliable estimate of the gradient direction under stochastic data augmentations,

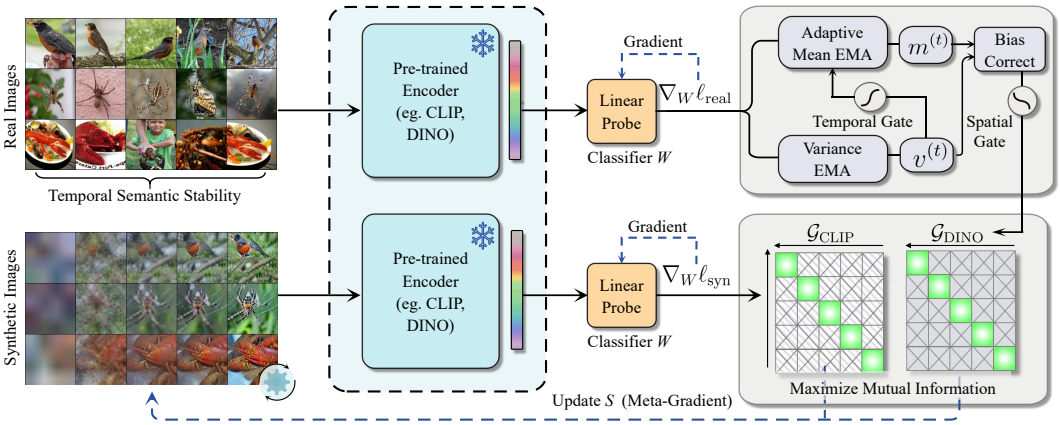


Figure 3: **Overview of Temporal Prototype Alignment.** (i): To reconstruct stable prototypes $\mathcal{G}$, an Adaptive Temporal Momentum module decouples semantic signals and imposes implicit multi-view regularization, using a temporal gate and spatial gate. (ii): $\mathcal{G}$ guides the synthesis of images. To ensure architecture-agnostic transferability, we perform Contrastive Consensus Alignment, maximizing the mutual information between $\nabla_{W_m} \ell_{\text{syn}}$ and $\mathcal{G}_m$ across heterogeneous backbones (e.g., CLIP, DINO) via a variational InfoNCE objective.

LGM employs a spatial Monte Carlo approximation by aggregating gradients across an ensemble of K concurrent views. Specifically, the synthetic loss is computed as:

$$\ell_{\text{syn}} = \frac{1}{K} \sum_{k=1}^{K} \text{CE}\left(W \phi(A_k(x_{\text{syn}})), y_{\text{syn}}\right), \tag{2}$$

with an analogous formulation for ℓ_{real}, where $A_k(\cdot)$ denotes differentiable data augmentations (e.g., RandomResizedCrop). In the IPC=1 regime, the multi-view objective is used to reduce gradient stochasticity and enforce consistency, thereby preventing view-specific shortcuts.

Limitations. Despite its effectiveness, LGM suffers from severe computational overhead. Processing K views (typically K=10) per iteration inflates the batch size to $2 \times C \times K$, leading to a linear growth in memory and I/O overhead. For instance, distilling ImageNet-1K requires **12 hours of training on 4×H200 GPUs**. Crucially, naively reducing K is not a viable solution, as it both introduces gradient stochasticity and weakens the multi-view consistency constraint.

These limitations motivate a paradigm shift from costly spatial averaging to efficient temporal integration. Figure 3 shows that our approach addresses these challenges through two synergistic modules: (i) Adaptive Temporal Momentum that stabilizes stochastic single-view gradients into robust distillation prototypes, and (ii) Contrastive Consensus Alignment that distills architecture-agnostic patterns by maximizing mutual information across heterogeneous models.

3.2 Adaptive Temporal Momentum

To circumvent the bottleneck of LGM, we reduce the augmentation ensemble size from K=10 to K=1. Thus, $\nabla_W \ell_{\text{real}}$ becomes highly stochastic and view-dependent. Departing from the traditional use of gradient history for acceleration, we reinterpret this temporal sequence as a mechanism to enforce multi-view consistency over time.

Specifically, we aim to reconstruct a stable distillation prototype $\mathcal{G}$ that captures the invariant semantic signal shared across random augmentations. This formulation entails two core challenges: (i) *Temporal gating:* the update trajectory must be gated by signal quality, as blind accumulation risks embedding outliers into the prototype. (ii) *Feature selectivity*: the filter must explicitly distinguish stable semantic signals from transient artifacts, suppressing feature dimensions dominated by high-frequency noise.

To this end, we propose Adaptive Temporal Momentum, a gating mechanism that explicitly decouples semantic signal from stochastic noise. Let m and v in $\mathbb{R}^{C \times F}$ denote the first- and second-order moment estimates of $\nabla_W \ell_{\text{real}}$. We modulate the accumulation of these moments via a soft temporal gate $\mu \in \mathbb{R}^{C \times F}$. At step t, the moments $m^{(t)}$ and $v^{(t)}$ update as

$$m^{(t)} = \mu^{(t)} \odot m^{(t-1)} + \left(1 - \mu^{(t)}\right) \odot \nabla_W \ell_{\text{real}}^{(t)}, \tag{3}$$

and

$$v^{(t)} = \beta v^{(t-1)} + (1-\beta)\left(\nabla_W \ell_{\text{real}}^{(t)}\right)^2, \tag{4}$$

where $\odot$ denotes the element-wise product and β is a fixed variance decay rate. Bounded by $\mu_{\min}$ and $\mu_{\max}$, the temporal gate $\mu^{(t)}$ is modulated by the local Signal-to-Noise Ratio rather than the static decay used in standard EMA:

$$\mu^{(t)} = \mu_{\min} + (\mu_{\max} - \mu_{\min}) \tanh\left(\frac{\sqrt{v^{(t)}}}{|m^{(t-1)}| + \epsilon}\right), \tag{5}$$

where ϵ is a small constant, and $\sqrt{v^{(t)}}/|m^{(t-1)}|$ approximates the inverse SNR. This operates as a Kalman-like Gain, which attenuates updates when transient noise dominates, while accelerating adaptation to consistent semantic trends.

To eliminate the initialization bias inherent to exponential moving averages, we compute the bias-corrected moment estimates:

$$\hat{m}^{(t)} = \frac{m^{(t)}}{1 - \prod_{i=1}^{t} \mu^{(i)}}, \quad \hat{v}^{(t)} = \frac{v^{(t)}}{1 - \beta^t}. \tag{6}$$

Subsequently, denoting the total optimization steps as T, we derive the distillation prototype trajectory $\mathcal{G} = \{\mathcal{G}^{(t)}\}_{t=1}^{T}$ via adaptive inverse-variance reweighting:

$$\mathcal{G}^{(t)} = \left(\frac{1}{\sqrt{\hat{v}^{(t)}} + \epsilon}\right) \odot \hat{m}^{(t)}, \tag{7}$$

where the element-wise factor acts as an adaptive spatial gate that suppresses noise-dominated dimensions while amplifying features with stable semantic consistency. Unlike conventional momentum that controls *how fast* optimization proceeds, this second-order moment determines *what* to learn by selectively filtering semantically reliable dimensions. Figure 4 reports the cosine similarity between consecutive gradient steps: real gradients with $K = 1$ fluctuate sharply under single-view noise, whereas our temporal integration reconstructs a stable distillation prototype that guides synthetic data toward consistent features.

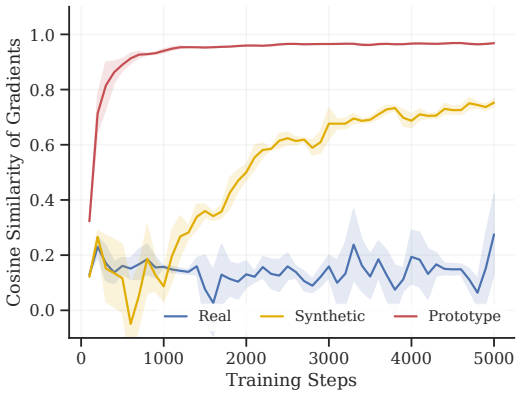


Figure 4: **Gradient Angular Stability Analysis.** Real gradients exhibit high stochasticity with similarity scores near 0.1. In contrast, our prototype maintains high temporal stability, providing a robust signal for synthesis.

3.3 Contrastive Consensus Alignment

Robust distilled data should capture invariant knowledge transferable across heterogeneous models. Hence, we propose distilling consensus knowledge from an ensemble of M diverse pre-trained backbones $\{\phi_m\}_{m=1}^{M}$, where each backbone features a dimension F_m and a corresponding model-specific linear head $W_m \in \mathbb{R}^{C \times F_m}$. Instead of relying on geometric matching, we frame the distillation process from an information-theoretic perspective.

Formally, we formulate this as maximizing the mutual information $\mathcal{I}(\cdot;\cdot)$ between the synthetic gradients of ϕ_m and its distillation prototypes $\mathcal{G}_m$ across heterogeneous feature spaces. We compute the objective with respect to W_m:

$$x_{\text{syn}}^{*} = \arg\max_{x_{\text{syn}}} \sum_{m=1}^{M} \mathcal{I}\left([\nabla_{W_m} \ell_{\text{syn}}]_y; \mathcal{G}_{m,y}\right), \tag{8}$$

where x_{syn}^{*} serves as a shared information bottleneck. Since x_{syn} is the only learnable parameter, it is driven by joint mutual information maximization to converge upon the semantic intersection of diverse architectural frameworks. Since directly optimizing mutual information is computationally intractable, we follow established works [48–50] to maximize a variational lower bound. Specifically, the mutual information is bounded by the InfoNCE loss as follows:

$$\sum_{m=1}^{M} \mathcal{I}\left([\nabla_{W_m} \ell_{\text{syn}}]_y; \mathcal{G}_{m,y}\right) \geq M \log(C) - \sum_{m=1}^{M} \mathcal{L}_{\text{NCE}}^{m}. \tag{9}$$

Table 1: **Linear Probing Accuracy (%) on ImageNet-Scale Datasets with 1 Image Per Class.** Notably, our single-model approach (i.e., Ours$_{[C/D/M]}$) achieves comparable performance to LGM while significantly reducing compute costs. MGD^3 and CaO2 are DiT-based generative dataset distillation baselines.

Train Set (1 Img/Cls)	ImageNet-100				ImageNet-1K			
	CLIP	DINOv2	MoCo-v3	Average	CLIP	DINOv2	MoCo-v3	Average
Full Dataset	92.5±0.0	95.2±0.1	89.4±0.3	92.4±0.2	78.7±0.0	83.0±0.0	76.5±0.0	79.4±0.0
Neighbors	67.8±0.3	86.0±0.2	77.1±0.1	76.9±0.2	38.8±0.1	67.7±0.1	56.4±0.0	54.3±0.1
Centroids	77.1±0.1	86.9±0.3	77.7±0.1	80.6±0.2	53.9±0.0	69.5±0.1	57.4±0.0	60.3±0.1
Random	56.6±1.6	74.8±2.8	61.4±2.6	64.3±2.4	31.7±0.5	50.3±0.5	38.8±0.6	40.3±0.5
MGD^3 [55]	68.5±0.2	82.1±0.1	72.1±0.2	74.2±0.2	45.4±0.1	62.4±0.1	50.9±0.1	52.9±0.1
CaO2 [56]	68.7±0.2	83.0±0.1	75.0±0.2	75.6±0.2	47.7±0.1	65.1±0.1	53.6±0.1	55.4±0.1
LGM [7]	84.9±0.1	91.5±0.1	83.4±0.1	86.6±0.1	63.0±0.0	75.0±0.1	63.2±0.0	67.1±0.0
Ours$_{[C/D/M]}$	86.7±0.2	92.2±0.1	85.2±0.1	88.0±0.1	63.8±0.1	75.6±0.1	63.6±0.1	67.7±0.1
Ours$_{[C+D+M]}$	**87.4**±0.1	**92.4**±0.1	**86.5**±0.1	**88.8**±0.1	**64.8**±0.1	**76.5**±0.1	**64.3**±0.2	**68.5**±0.1

Table 2: **Cross-Architecture Linear Probing (%) and Memory Overhead on ImageNet-100.** Our method achieves superior generalization across 9 unseen architectures with 10x memory reduction compared to LGM.

Distill Model	GPU Mem.	Self-Supervised Models				Supervised Models					Average
		EVA-02	iBOT	SimMIM	BEiT	ResNet50	ConvNext	Swin-T	MobileV3	EffNet	
LGM trained on the 4 × RTX 4090											
C	81G	83.6±0.3	75.6±0.0	92.1±0.1	87.8±0.1	69.8±0.4	91.3±0.3	89.3±0.4	73.0±0.0	76.8±0.2	82.1±0.2
D	81G	85.2±0.1	83.4±0.3	94.5±0.0	94.9±0.1	88.7±0.3	94.2±0.3	94.8±0.1	80.2±0.3	85.1±0.1	89.0±0.2
Ours trained on the RTX 4090											
C	7G	83.6±0.4	79.5±0.3	95.2±0.2	89.5±0.1	78.2±0.5	93.5±0.2	93.4±0.0	77.0±0.0	78.0±0.4	85.3±0.3
D	7G	86.6±0.0	84.4±0.4	96.4±0.3	95.0±0.5	86.9±0.2	96.4±0.3	94.6±0.4	79.6±0.3	83.9±0.4	89.3±0.3
C+D	8G	88.3±0.4	87.5±0.0	98.0±0.1	95.0±0.1	**91.4**±0.3	**97.4**±0.1	95.9±0.1	86.8±0.1	86.5±0.3	92.0±0.2
C+D+M	8G	**88.6**±0.2	**87.7**±0.1	**98.6**±0.1	**95.6**±0.2	**91.4**±0.0	97.3±0.1	**96.0**±0.1	**88.3**±0.4	**87.4**±0.1	**92.3**±0.1

Accordingly, our objective is to minimize the summation of model-specific InfoNCE losses. We define the class-matched prototype $\mathcal{G}_{m,y}$ as the positive for row $[\nabla_{W_m}\ell_{\text{syn}}]_y$, while the prototypes from other classes $\{\mathcal{G}_{m,j}\}_{j\neq y}$ serve as negative samples. The contrastive formulation encourages the synthetic gradients to capture class-discriminative information by aligning with their corresponding prototypes. The model-specific InfoNCE loss is defined as:

$$\mathcal{L}^m_{\text{NCE}} = -\mathbb{E}_y \left[\log \frac{\exp\left(\text{sim}([\nabla_{W_m}\ell_{\text{syn}}]_y, \mathcal{G}_{m,y})/\tau\right)}{\sum_{j=1}^{C} \exp\left(\text{sim}([\nabla_{W_m}\ell_{\text{syn}}]_y, \mathcal{G}_{m,j})/\tau\right)} \right]. \tag{10}$$

where y indexes a class row, sim is cosine similarity, and τ is the temperature. We achieve consensus distillation via joint InfoNCE optimization. For example, utilizing CLIP captures semantic context, while DINO preserves structural details, ensuring x_{syn} captures shared patterns across feature spaces.

4 Experiments

Datasets. We evaluate the proposed method on four benchmark datasets spanning both large-scale classification and specialized fine-grained recognition. Specifically, we use ImageNet-1K [51] and its subset ImageNet-100 for large-scale validation, and Food-101 [52] and Flowers-102 [53] to assess fine-grained performance.

Implementation Details. All experiments are conducted on NVIDIA RTX 4090 GPUs at a resolution of 224×224. We employ the Adam optimizer [54] to update the synthetic images for 5,000 iterations. We fix $\mu_{\min}$=0.5 and $\mu_{\max}$=0.99 across all datasets. We utilize the frozen ViT-B architectures of CLIP, DINOv2, and MoCo-v3. Unless otherwise specified, reported means and standard deviations summarize five runs. More details are provided in the Appendix.

Evaluation and Baselines. We evaluate distilled images via linear probing and 1-shot black-box transfer: the former trains a classifier on frozen source features, while the latter uses distilled images as class prototypes for training-free nearest-neighbor inference across nine unseen architectures.

We compare our method with selection-based baselines, LGM [7], and two DiT-based generative dataset distillation baselines, MGD^3 [55] and CaO2 [56]. The selection baselines include random sampling, class centroids, and nearest neighbors, while LGM represents the current state-of-the-art for

Table 3: **Linear Probing Accuracy (%) on Fine-Grained Datasets with 1 Image Per Class.** Our cross-model consistency method yields significant performance gains over LGM, notably achieving an improvement of 7.2% on Food-101. For fine-grained datasets, ImageNet-1K class-conditional DiT samples from MGD^3 and CaO2 are mapped to target classes by CLIP text similarity before linear probing.

Train Set (1 Img/Cls)	Flowers-102				Food-101			
	CLIP	DINOv2	MoCo-v3	Average	CLIP	DINOv2	MoCo-v3	Average
Full Dataset	93.4±0.1	99.7±0.0	82.2±0.1	91.8±0.1	91.9±0.0	92.7±0.1	78.3±0.0	87.6±0.0
Neighbors	69.4±0.8	99.4±0.2	48.6±1.8	72.5±0.9	58.8±1.3	73.8±0.4	25.2±1.3	52.6±1.0
Centroids	77.5±0.7	99.4±0.1	48.2±2.6	75.0±1.1	74.1±0.6	75.6±0.1	22.7±2.0	57.5±0.9
Random	67.9±1.1	98.8±0.4	36.8±2.2	67.8±1.2	48.5±1.0	52.8±1.9	13.5±1.7	38.3±1.5
MGD^3 [55]	6.4±0.2	7.2±0.1	6.6±0.3	6.7±0.2	5.0±0.2	3.8±0.1	3.5±0.3	4.1±0.2
CaO2 [56]	6.9±0.3	7.1±0.2	6.8±0.4	6.9±0.2	5.6±0.3	4.6±0.2	3.5±0.4	4.6±0.3
LGM [7]	79.5±1.2	99.6±0.0	56.4±3.1	78.5±1.5	78.7±1.6	**83.7±0.3**	31.6±3.2	64.7±1.7
Ours$_{[C/D/M]}$	79.8±0.8	**99.7±0.1**	57.1±1.3	78.9±0.6	80.7±0.8	83.3±0.2	49.1±1.2	71.0±0.7
Ours$_{[C+D+M]}$	**84.3±0.2**	**99.7±0.1**	**61.8±1.0**	**81.9±0.5**	**82.6±0.6**	83.6±0.2	**49.4±0.7**	**71.9±0.5**

Table 4: **Black-Box 1-Shot Transfer Performance (%) and Training Time on ImageNet-100.** Distilled images are used as prototypes for training-free inference. 'Time' denotes total distillation hours.

Distill Model	GPU Time	Self-Supervised Models				Supervised Models					Average
		EVA-02	iBOT	SimMIM	BEiT	ResNet50	ConvNext	Swin-T	MobileV3	EffNet	
LGM trained on the 4 × RTX 4090											
C	3.6h	71.7±0.2	69.5±0.3	77.8±0.1	77.9±0.2	34.0±0.4	66.3±0.2	71.5±0.1	63.8±0.3	70.1±0.2	66.9±0.2
D	3.6h	80.4±0.1	84.1±0.2	88.5±0.3	92.8±0.2	61.5±0.3	85.0±0.1	83.4±0.2	73.2±0.3	77.7±0.2	80.7±0.2
Ours trained on the RTX 4090											
C	0.8h	78.6±0.1	76.1±0.2	81.3±0.3	82.0±0.1	46.5±0.4	73.6±0.2	76.6±0.1	70.3±0.3	76.3±0.2	73.5±0.2
D	0.8h	82.1±0.2	83.0±0.1	90.1±0.2	**94.5±0.1**	63.1±0.3	85.3±0.1	83.8±0.2	74.0±0.0	80.2±0.2	81.8±0.2
C+D	1.7h	85.7±0.1	85.7±0.2	91.3±0.2	93.6±0.3	73.2±0.2	86.7±0.1	84.3±0.3	80.5±0.2	85.8±0.2	85.2±0.2
C+D+M	2.6h	**86.5±0.1**	**88.1±0.2**	**93.8±0.1**	93.9±0.2	**81.2±0.3**	**89.5±0.1**	**88.5±0.2**	**84.2±0.2**	**89.4±0.1**	**88.4±0.2**

distillation in frozen feature spaces. MGD^3 and CaO2 synthesize class-conditional training images using DiT-based generators instead of selecting real samples. We also report full-dataset performance as an empirical upper bound. Details of all baseline settings are deferred to the Appendix.

4.1 Performance on Large-Scale Benchmarks

We evaluate the performance of our method on ImageNet-100 and ImageNet-1K under the IPC=1. As reported in Table 1, the proposed consensus-driven variant, Ours$_{[C+D+M]}$, establishes new state-of-the-art results across both scales. On ImageNet-100, our full method achieves an average accuracy of 88.8%, exceeding LGM by 2.2 percentage points. This performance leadership extends to the full ImageNet-1K dataset, where Ours$_{[C+D+M]}$ reaches 68.5%, surpassing LGM by 1.4 points. Qualitatively, the distillation visualizations in Figure 5 demonstrate that our synthesized images for both ImageNet scales are semantically coherent, capturing salient textural and structural features.

As shown in Figure 1, our method provides a stronger accuracy-efficiency trade-off than LGM. Specifically, LGM requires 3.6 hours of distillation time and a peak GPU memory of 81 GB to reach 84.9% accuracy with CLIP on ImageNet-100. In contrast, the single-backbone variant reported in Table 1, Ours$_{[C/D/M]}$, reaches 86.7% CLIP accuracy while requiring only 0.8 hours and 7 GB memory. Scaling to three backbones further raises Ours$_{[C+D+M]}$ to 87.4% CLIP accuracy in 2.6 hours with only 8 GB memory, making high-performance distillation feasible on consumer-grade hardware.

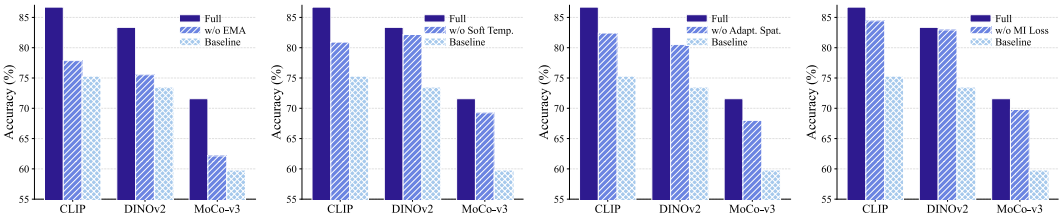
4.2 Cross-Architecture Transferability

A key objective of our method is to distill architecture-agnostic images that ensure cross-model generalization. We validate this capability through a comprehensive evaluation protocol comprising linear probing for cross-model adaptation and 1-shot inference for black-box transfer. As shown in Table 2, although distilled data is inherently limited by source model quality, single-model baselines suffer additional degradation from feature misalignment. In contrast, the proposed consensus method, Ours$_{[C+D+M]}$, achieves consistently higher accuracy across heterogeneous architectures while retaining practical efficiency. Our method completes distillation in just 2.6 hours on a single RTX 4090.



(a) ImageNet-1K (b) ImageNet-100 (c) Food-101 (d) Flowers-102

Figure 5: **Qualitative Results of Distillation Prototypes.** These images capture highly condensed, class-discriminative semantics.



(a) EMA Update (b) Soft Temporal Gate (c) Adaptive Spatial Gate (d) MI Objective

Figure 6: **Ablation Studies on ImageNet-100**. We evaluate the linear probing accuracy on the teacher (CLIP) and cross-model backbones (DINOv2, MoCo-v3). "Full" is our complete framework, while "Baseline" denotes the performance without any proposed components.

The superiority of our approach is most evident in the 1-shot results presented in Table 4. Images distilled from a single backbone often suffer severe degradation when transferred to dissimilar architectures, a failure mode best exemplified by the collapse of LGM-C to 34.0% accuracy on ResNet-50. In sharp contrast, Ours$_{[C+D+M]}$ achieves 81.2% on the same target, effectively bridging the representational gap between transformers and convolutional networks. Averaged across nine unseen models, our framework attains 88.4%, outperforming the LGM-C and LGM-D baselines. As further visualized in the radar charts of Figures 2a and 2b, these results suggest that our method captures visual features that transfer across the tested models.

4.3 Fine-Grained Visual Classification

Unlike standard classification task, fine-grained datasets such as Food-101 and Flowers-102 demand capturing subtle discriminative features to distinguish closely related categories. As indicated in Table 3, Ours$_{[C+D+M]}$ achieves an average accuracy of 71.9% on Food-101, exceeding LGM by 7.2 percentage points. This advantage is most pronounced in recovering failure modes, where we boost the MoCo-v3 baseline from 31.6% to 49.4%. On Flowers-102, Ours$_{[C+D+M]}$ reaches a new state-of-the-art average accuracy of 81.9%, while the DINOv2 source-matched variant matches the empirical ceiling at 99.7%. Qualitatively, the distilled images in Figure 5 demonstrate that prototypes encapsulate the distinctive structural and textural patterns.

4.4 Ablation Study

We conduct comprehensive ablation studies on ImageNet-100 under IPC=1 to quantify the individual contribution of each core component. The distilled data are evaluated by linear probing on the source CLIP model, and their cross-architecture transferability is further assessed on unseen DINOv2 and MoCo-v3 backbones. As shown in Figure 6, removing temporal integration leads to a sharp performance drop across all models, confirming that single-view updates are dominated by transient noise. This establishes historical accumulation as a core mechanism for smoothing optimization trajectories and recovering invariant semantic signals.

Temporal-Spatial Integration. Figure 6 further validates the importance of both temporal and spatial adaptivity. Replacing the SNR-modulated soft temporal gate with a static decay consistently degrades performance, indicating that dynamically weighting updates is essential for decoupling persistent

Table 7: **Teacher Heterogeneity and Consensus Capacity on ImageNet-100.** We report results for adding heterogeneous teacher biases. * denotes 1-shot transfer accuracy. Gen. denotes the generative prior from MAE. ConvN, MobV3, and EffN denote ConvNeXt, MobileNetV3, and EfficientNet.

Config.	Bias	Mem.	CLIP	DINOv2	MoCo-v3	RN50*	ConvN*	MobV3*	EffN*
C+D+M	ViT	8.0GB	**87.4**	**92.4**	**86.5**	81.2	89.5	84.2	**89.4**
+MAE	Gen.	8.2GB	**87.4**	92.0	86.2	81.6	89.7	84.3	89.0
+ConvN	CNN	12.9GB	86.6	92.2	86.3	**82.3**	**90.7**	**84.4**	88.5

signal from timestep-dependent noise. Similarly, removing the adaptive spatial gate, especially inverse-variance reweighting, compromises transferability, with a notable drop on MoCo-v3. These results show that temporal accumulation, adaptive temporal gating, and spatial feature selectivity are complementary mechanisms for producing robust and transferable distilled prototypes.

Mutual Information Objective. Figure 6d shows that removing the InfoNCE-based objective compromises data quality. This supports class-wise contrastive matching between synthetic gradients and prototypes as an effective objective for cross-architecture transfer in this setting.

Model Combinations. Table 5 shows that the full triplet ensemble achieves the most robust generalization. Removing any single teacher leads to a performance drop on the corresponding architecture, indicating that fusing complementary teacher perspectives is essential for forming a consensus prototype that avoids overfitting to any single representation space. Although multi-teacher distillation increases FLOPs, TPA keeps the sequential C+D+M peak memory at only 8.0GB. As shown in Table 6, two RTX 4090 GPUs reduce C+D+M distillation from 156m to 109m and complete C+D in 52m with a 14.2GB joint footprint, whereas LGM's 81 GB footprint makes comparable commodity-GPU parallelization difficult.

Table 5: **Ablation of Teacher Components on ImageNet-100.** ✓ denotes that the teacher is used for distillation.

Distilled Models			Validation Accuracy (%)			
C	D	M	CLIP	DINOv2	MoCo-v3	Avg.
✓	✓		86.8	92.3	83.6	87.5
✓		✓	86.6	89.9	86.3	87.6
	✓	✓	82.1	92.3	86.2	86.8
✓	✓	✓	**87.4**	**92.4**	**86.5**	**88.8**

Table 6: **Empirical Compute Complexity on ImageNet-100.** Sequential runs teachers one by one. Distributed splits teacher/batch workloads across two RTX 4090 GPUs.

Setup	Strategy	GPU	Mem.	Time
C+D+M	Sequential	1×4090	**8.0GB**	156m
C+D	Distributed	2×4090	14.2GB	**52m**
C+D+M	Distributed	2×4090	14.2GB	**109m**

Consensus Capacity. Table 7 studies whether enlarging the teacher set beyond the ViT triplet further improves consensus distillation. Adding MAE introduces a generative prior with little memory overhead, but yields only marginal changes and slightly reduces DINOv2 and MoCo-v3 probing accuracy. Adding ConvNeXt brings a stronger convolutional texture bias and improves transfer to CNN targets, e.g., ResNet-50 increases from 81.2% to 82.3% and ConvNeXt from 89.5% to 90.7%. However, this gain comes with a higher memory cost and a small drop on ViT-oriented linear probing. These results show that consensus distillation is not simply improved by adding more teachers. Heterogeneous teachers may introduce useful architecture-specific priors, but overly broad consensus can also force conflicting signals into the same synthetic image. The triplet therefore offers the best performance-cost trade-off in our setting.

4.5 Additional Analysis

Increasing synthetic capacity. At IPC=5, TPA (C+D+M) reaches 89.1% on ImageNet-100 and 74.8% on Food-101, exceeding LGM by 2.0 and 3.6 percentage points (Table 8). The corresponding encoder-level results appear in Appendix Table 16. Compared with single-teacher TPA at the same IPC, consensus alignment improves accuracy by 1.7 points on both datasets, extending the multi-teacher gains in Section 4.4 to a larger synthetic set. Table 9 examines the effect of synthetic-set size within single-teacher TPA by fixing the training protocol and increasing IPC from 1 to 5. Accuracy improves across all three encoders by 0.78–1.28 points, averaging 0.97 points. The consistent gains across CLIP, DINOv2, and MoCo-v3 show that additional synthetic capacity benefits different pretrained representations within the same optimization budget.

Table 8: Higher-IPC performance. IPC=5 linear probing accuracy (%) averaged over CLIP, DINOv2, and MoCo-v3, reported as five-run means and standard deviations. LGM uses $K = 2$ and TPA uses $K = 1$. The single-teacher batch size is 128.

Method	ImageNet-100	Food-101
LGM	87.1 ± 0.3	71.2 ± 0.3
TPA (single)	87.4 ± 0.2	73.1 ± 0.3
TPA (C+D+M)	$\mathbf{89.1} \pm 0.1$	$\mathbf{74.8} \pm 0.2$
Gain over LGM (pp)	+2.0	+3.6

Table 9: Effect of synthetic-set size. ImageNet-100 linear probing accuracy (%) using the same encoder for distillation and evaluation. Both IPC settings use the full ATM estimator, batch size 256, $K = 1$, and 5,000 steps, with one evaluation per synthetic set.

Encoder	IPC=1	IPC=5	Δ (pp)
CLIP	86.52	87.80	+1.28
DINOv2	92.00	92.86	+0.86
MoCo-v3	85.84	86.62	+0.78
Mean	88.12	89.09	+0.97

Temporal prototype quality. Table 10 measures agreement between the temporal prototypes and an independent reference formed from 512 real-gradient observations with inverse-RMS weighting. Cosine similarity increases from 0.9074 in the early stage to 0.9757 in the middle stage and 0.9824 in the late stage. Scale-invariant MSE decreases from 0.1733 to 0.0480 and then to 0.0348, giving an early-to-late reduction of approximately 80%. The largest gains occur during the initial accumulation period, followed by smaller improvements as more observations are incorporated. Whereas Figure 4 measures agreement between consecutive updates, these metrics quantify fidelity to a reference that uses the same form of feature weighting as the prototype estimator. Together, the two analyses show that ATM combines temporal consistency with increasing reference fidelity, supporting its use as a supervision target for synthetic-image optimization.

Table 10: Evolution of prototype quality. One ImageNet-100/CLIP trajectory with batch size 256. Metrics average the listed snapshots and all 100 classes against the independent inverse-RMS reference.

Stage (optimization steps)	Cosine $\uparrow$	$1 - \text{cosine} \downarrow$	SI-MSE $\downarrow$
Early (100/500)	0.9074	0.0926	0.1733
Middle (1,000/2,000)	0.9757	0.0243	0.0480
Late (3,000/4,000/5,000)	0.9824	0.0176	0.0348

Limitations and future directions. TPA relies on class labels and frozen teachers to define consistent gradient targets throughout distillation. Extending the method to unlabeled data requires learning class correspondences alongside the synthetic images, while evolving teachers introduce gradient drift that the temporal estimator must track. These extensions motivate studying how adaptive gating and inverse-RMS weighting interact with changing supervision. The teacher trade-offs in Table 7 also motivate automatic selection of complementary pretrained models. Comparing class-level relations on training data could provide a common basis for selecting teachers with different feature dimensions. Such selection would aim to balance transfer across target architectures with distillation cost, building on the complementary supervision provided by the current teacher triplet.

5 Conclusion

In this paper, we introduced Temporal Prototype Alignment, a scalable dataset distillation framework that replaces computationally expensive spatial ensembling with temporal accumulation. Adaptive Temporal Momentum (ATM) constructs stable teacher-local prototypes, and class-wise contrastive alignment transfers their complementary supervision to a shared synthetic dataset. In the ImageNet-100 comparison with LGM, single-teacher TPA reduces peak GPU memory from 81 GB to 7 GB, a reduction of approximately 91%, while improving accuracy. TPA (C+D+M) further improves transfer to unseen target models, including convolutional networks such as ResNet-50 and ConvNeXt and vision transformers (Tables 2 and 4). These results support temporal prototype alignment as an efficient approach to transferable dataset distillation with frozen encoders.

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A Appendix Overview

The appendix is organized to provide implementation details, theoretical justification, supplementary diagnostics, and qualitative evidence supporting the main paper. Appendix C describes the full experimental protocol, including the distillation setup, teacher and target model suites, data preprocessing, linear probing evaluation, and training-free 1-shot transfer evaluation. Appendix D.1 gives the derivation of the variational lower bound used by the mutual-information objective, clarifying how the InfoNCE loss encourages cross-model consensus between synthetic gradients and temporal prototypes. Appendix D.2 then reports supplementary diagnostics, including spatial-view sensitivity, batch-size fairness, teacher scaling under extreme compression, extended black-box transfer, I/O profiling, and the relation to train-from-scratch dataset distillation. Appendix D.8 studies the sensitivity of the temporal gate thresholds, while Appendix D.9 evaluates the method under a larger IPC=5 distillation budget. Finally, Appendix E presents additional synthesized-image visualizations for both single-teacher and multi-teacher settings, including ImageNet-scale and fine-grained datasets.

B Impact Statement

This work contributes to the field of Effective Dataset Distillation by introducing a memory-efficient framework that operates on frozen features. A primary societal impact of our research is the reduction of computational resources required for training and data synthesis, which aligns with the goals of Green AI and sustainable computing. By making high-quality synthetic data accessible with limited hardware resources, we hope to facilitate broader participation in data-efficient learning research. We do not foresee immediate negative societal consequences, though general considerations regarding the use of synthetic data apply.

C Experimental Details

C.1 Implementation Details

We implement Temporal Prototype Alignment in PyTorch and run the reported distillation experiments on NVIDIA RTX 4090 GPUs. Unless otherwise specified, each synthetic set is optimized for $T = 5{,}000$ iterations with Adam [54] using learning rate $\eta = 2 \times 10^{-3}$. The source teachers are frozen throughout distillation. The default consensus configuration uses CLIP ViT-B/16, DINOv2 ViT-B/14, and MoCo-v3 ViT-B/16.

Synthetic images are represented with a multi-scale pyramid parameterization. A new resolution band is activated every 200 iterations, and each band is initialized with $\mathcal{N}(0, 1)$. The rendered synthetic image is obtained by summing the bilinearly upsampled pyramid bands and applying a smooth clamp:

$$I_{\text{syn}} = \frac{1}{2} + \frac{1}{2} \tanh\left(\sum_{l=0}^{L} \texttt{Upsample}(P_l) \right). \tag{11}$$

Real images are resized to 256×256 and randomly cropped to 224×224 during distillation. The differentiable augmentation pipeline applies random resized crop, horizontal flip with probability 0.5, and Gaussian perturbations $\mathcal{N}(\mu = 0.5, \sigma = 0.2)$. TPA uses one differentiable view per iteration ($K = 1$), while LGM baselines use their reported spatial-view settings.

C.2 Model Suite and Evaluation

We evaluate each distilled set with both source backbones and unseen target models. The target suite spans masked-image-modeling transformers (iBOT, BEiT v2, SimMIM, EVA-02), hierarchical transformers (Swin), and convolutional architectures (ResNet-50, ConvNeXt, EfficientNet, MobileNetV3). Real-image baselines follow the LGM protocol [7]. For distillation, the default batch size is 256 for single-teacher runs and 128 for multi-teacher runs. The original single-teacher IPC=5 results in Table 16 also use 128, whereas Table 9 uses 256 for both IPC values.

Linear probing. For each evaluation backbone, we freeze the pretrained feature extractor and use the distilled images as the only training set for a newly initialized linear classifier. The classifier is

trained from scratch on top of fixed features, without updating the backbone or using any real training images. Test images are resized to 256 and center-cropped to 224×224 before feature extraction. **1-shot inference.** For the non-parametric evaluation, each distilled image is encoded by the target backbone and treated as the prototype of its corresponding class. Each test image is assigned to the class whose prototype has the highest cosine similarity, without training a target-specific classifier or performing any target-backbone adaptation.

D Additional Results

D.1 Derivation of the Variational Lower Bound

Condition on a fixed head and prototype bank, with classes sampled uniformly. Assume distinct class prototypes, so $H(\boldsymbol{c}) = \log C$. The bound below characterizes the class information captured by synthetic gradient rows relative to their corresponding teacher-local prototypes. Let $\boldsymbol{v} = [\nabla_{W_m} \ell_{\text{syn}}]_y$ denote the synthetic gradient and $\boldsymbol{c} = \mathcal{G}_{m,y}$ denote the corresponding class-matched prototype. The set $\{\mathcal{G}_{m,j}\}_{j \neq y}$ represents the $C-1$ negative prototypes. The variational probability $q_m(\boldsymbol{c} \mid \boldsymbol{v})$ is defined as:

$$q_m(\boldsymbol{c} \mid \boldsymbol{v}) = \frac{\exp(\text{sim}(\boldsymbol{v}, \boldsymbol{c})/\tau)}{\sum_{j=1}^{C} \exp(\text{sim}(\boldsymbol{v}, \mathcal{G}_{m,j})/\tau)} \tag{12}$$

The derivation proceeds as follows:

$$\begin{aligned}
\mathcal{L}_{\text{NCE}}^{m} &= -\mathbb{E}_{(\boldsymbol{v},\boldsymbol{c})\sim p(\boldsymbol{v},\boldsymbol{c})}\Big[\log q_m(\boldsymbol{c} \mid \boldsymbol{v})\Big] \\
&= \mathbb{E}_{\boldsymbol{v}}\left[\sum_{\boldsymbol{c}\in\{\mathcal{G}_{m,1},\ldots,\mathcal{G}_{m,C}\}} p(\boldsymbol{c} \mid \boldsymbol{v})\Big(-\log q_m(\boldsymbol{c} \mid \boldsymbol{v})\Big)\right] \\
&= \mathbb{E}_{\boldsymbol{v}}\left[\sum_{\boldsymbol{c}} p(\boldsymbol{c} \mid \boldsymbol{v})\Big(-\log q_m(\boldsymbol{c} \mid \boldsymbol{v})\Big)\right] \\
&= \mathbb{E}_{\boldsymbol{v}}\left[\sum_{\boldsymbol{c}} p(\boldsymbol{c} \mid \boldsymbol{v})\Big(-\log p(\boldsymbol{c} \mid \boldsymbol{v})\Big) + \sum_{\boldsymbol{c}} p(\boldsymbol{c} \mid \boldsymbol{v}) \log \frac{p(\boldsymbol{c} \mid \boldsymbol{v})}{q_m(\boldsymbol{c} \mid \boldsymbol{v})}\right] \\
&= \mathbb{E}_{\boldsymbol{v}}\left[\sum_{\boldsymbol{c}} p(\boldsymbol{c} \mid \boldsymbol{v})\Big(-\log p(\boldsymbol{c} \mid \boldsymbol{v})\Big)\right] + \mathbb{E}_{\boldsymbol{v}}\left[\sum_{\boldsymbol{c}} p(\boldsymbol{c} \mid \boldsymbol{v}) \log \frac{p(\boldsymbol{c} \mid \boldsymbol{v})}{q_m(\boldsymbol{c} \mid \boldsymbol{v})}\right] \\
&= H(\boldsymbol{c} \mid \boldsymbol{v}) + \mathbb{E}_{\boldsymbol{v}}\Big[\text{KL}\big(p(\cdot \mid \boldsymbol{v}) \,\|\, q_m(\cdot \mid \boldsymbol{v})\big)\Big] \\
&\geq H(\boldsymbol{c} \mid \boldsymbol{v}) \\
&= H(\boldsymbol{c}) - \mathcal{I}(\boldsymbol{v}; \boldsymbol{c}) \\
&= \log(C) - \mathcal{I}(\boldsymbol{v}; \boldsymbol{c})\,.
\end{aligned} \tag{13}$$

By rearranging the inequality, we validate the variational lower bound presented in the main paper:

$$\sum_{m=1}^{M} \mathcal{I}\left([\nabla_{W_m} \ell_{\text{syn}}]_y; \mathcal{G}_{m,y}\right) \geq M \log(C) - \sum_{m=1}^{M} \mathcal{L}_{\text{NCE}}^{m}. \tag{14}$$

Hence, jointly minimizing $\mathcal{L}_{\text{NCE}}^{m}$ maximizes the variational lower bound, enforcing x_{syn} as a shared information bottleneck that distills semantic intersections across heterogeneous architectures.

D.2 Sensitivity to Spatial Ensemble Size

We further analyze how the number of differentiable spatial views affects LGM under the single-teacher ImageNet-100 setting. This study is intended to clarify the role of spatial ensembling in gradient-based distillation and to contrast it with the temporal aggregation mechanism used by TPA. As shown in Table 11, LGM benefits substantially from increasing K: larger spatial ensembles

improve both source-model accuracy and cross-model transfer, especially on MoCo-v3. However, when K is reduced to 1, its transfer performance drops sharply, indicating that single-view gradient estimates are highly sensitive to augmentation-induced variance.

Table 11: **Sensitivity to Spatial Ensemble Size on ImageNet-100.** We report linear probing accuracy (%) after CLIP-teacher distillation, evaluated on CLIP, DINOv2, and MoCo-v3. Increasing K averages more spatial views in LGM, while TPA uses $K = 1$ and accumulates information temporally.

Method	K	CLIP	DINOv2	MoCo-v3	Average
LGM	1	77.9	71.7	44.2	64.6
LGM	2	82.3	78.7	50.4	70.5
LGM	3	83.6	78.0	53.4	71.7
LGM	10	84.9	84.5	60.1	76.5
TPA	1	**86.7**	**83.2**	**70.5**	**80.1**

These results highlight a key distinction between the two strategies. LGM improves stability by averaging gradients over simultaneous spatial views, which increases the per-iteration computational cost as K grows. In contrast, TPA keeps $K = 1$ and reconstructs stable supervision through temporal prototype accumulation. The stronger $K = 1$ performance of TPA therefore suggests that temporal integration can provide an effective alternative to large spatial ensembles, improving scalability while preserving cross-backbone transferability.

D.3 Batch-Size Analysis for Single-Teacher Distillation

We additionally study how the effective batch size affects single-teacher distillation on ImageNet-100. This analysis complements the K-sensitivity study by separating two sources of supervision density: the number of spatial views per sample and the number of real samples processed per iteration. LGM uses a large batch together with multiple spatial views, which provides a strong per-iteration estimate but incurs substantial memory cost. TPA instead uses temporal prototype accumulation, allowing us to increase the real-image batch size while keeping the memory footprint small.

Table 12: **Batch-Size and Memory Analysis on ImageNet-100.** We report source-matched linear probing accuracy (%). Each column uses its own teacher for distillation and evaluation. TPA maintains low GPU memory usage as batch size increases, without requiring large spatial ensembles.

Method	Batch	GPU Mem.	CLIP	DINOv2	MoCo-v3	Avg.
LGM	1000	81.2GB	84.9	91.5	83.4	86.6
TPA	256	7.0GB	**86.7**	92.2	85.2	88.0
TPA	512	7.1GB	86.2	**92.4**	**85.7**	**88.1**

The results show that increasing the TPA batch size from 256 to 512 yields similar average accuracy, suggesting that the method is not highly sensitive to this range once the temporal prototype estimates are stable. More importantly, both TPA configurations remain substantially more memory-efficient than the LGM single-teacher setting while achieving higher average accuracy. This supports the view that TPA improves the efficiency of the supervision estimate through temporal aggregation rather than relying on a large per-iteration spatial sampling budget.

D.4 Teacher Scaling under Extreme Compression

We additionally study whether using a stronger teacher backbone directly translates into better distilled prototypes under the IPC=1 constraint. This analysis complements the main results by separating two factors: the representational strength of an individual teacher and the capacity of a single synthetic prototype to absorb information from multiple teachers. As shown in Table 13, replacing DINOv2 with DINOv3 improves the single-teacher setting, suggesting that stronger teacher representations can provide more informative supervision.

Table 13: **Teacher Scaling under IPC=1 on ImageNet-100.** We report linear probing accuracy (%) across evaluation backbones. Stronger teachers improve single-teacher distillation, while gains become smaller in the multi-teacher setting due to the limited information capacity of one synthetic image per class.

Configuration	Setup Detail	CLIP	DINOv2	MoCo-v3	DINOv3	Avg.
Single Teacher	DINOv2	78.5	92.2	81.3	89.9	85.5
Single Teacher	DINOv3	81.6	91.1	83.1	92.2	87.0
Triplet Teacher	C+DINOv2+M	87.4	**92.4**	**86.5**	90.3	89.2
Triplet Teacher	C+DINOv3+M	**87.9**	91.1	**86.5**	**92.3**	**89.5**

The multi-teacher results show a different trend. Although DINOv3 improves its own evaluation column and slightly increases the average accuracy, the gain over the CLIP+DINOv2+MoCo-v3 triplet remains modest. This suggests that, under IPC=1, performance is not determined solely by teacher strength. It is also constrained by the finite capacity of a single prototype and by how complementary the teacher signals are. We therefore use the CLIP+DINOv2+MoCo-v3 consensus as our primary setting, as it provides a strong cost–performance trade-off while preserving broad cross-backbone transferability.

D.5 Extended Black-Box 1-Shot Transfer

We provide an extended black-box transfer evaluation to examine whether the distilled prototypes remain useful beyond the teacher backbones used during distillation. In this setting, each synthetic image is treated as a class prototype, and test samples are classified by nearest-neighbor cosine similarity under unseen target encoders. The target suite covers masked-image-modeling transformers, hierarchical transformers, and convolutional networks, allowing us to assess whether the distilled set captures broadly transferable semantics rather than features tied to a particular teacher family.

Table 14: **Extended Black-Box 1-Shot Transfer on ImageNet-100.** We report 1-shot nearest-prototype accuracy (%) on unseen target backbones. TPA achieves stronger average transfer while requiring substantially less distillation time.

Method	Setup	Time	EVA	SimMIM	BEiT	RN50	ConvN	Swin	MobV3	Eff.	Avg.
LGM	C+D	6.7h	82.2	89.0	93.0	66.3	85.2	83.6	78.1	82.5	82.5
LGM	C+D+M	9.7h	83.0	90.5	93.5	74.2	88.2	85.8	81.6	85.3	85.3
TPA	C+D+M	2.6h	**86.5**	**93.8**	**93.9**	**81.2**	**89.5**	**88.5**	**84.2**	**89.4**	**88.4**

The results show that adding teachers improves LGM, but the resulting transfer performance still trails TPA despite the higher distillation cost. The gains of TPA are particularly pronounced on structurally different CNN targets such as ResNet-50 and EfficientNet, suggesting that temporal consensus does not merely optimize for the source teacher representations. Instead, it helps recover class-level prototypes that are more invariant across architectural families, which is important for black-box reuse of distilled datasets.

D.6 I/O Throughput and DataLoader Profiling

Beyond accuracy and GPU memory, we also profile the input pipeline to understand how reducing spatial redundancy affects end-to-end distillation efficiency. All measurements are collected on ImageNet-100 over 5,000 iterations using a PCIe NVMe SSD and 16 DataLoader workers. As shown in Table 15, replacing large spatial ensembles with $K = 1$ temporal integration substantially reduces DataLoader latency. Compared with the multi-teacher LGM setting, TPA reduces pure I/O load from 0.062s to 0.008s per iteration, corresponding to a $7.8\times$ reduction, and decreases total preparation time from 0.220s to 0.043s per iteration, corresponding to a $5.1\times$ reduction. Meanwhile, the effective throughput increases from 11.9 GB/s to 21.6 GB/s, yielding a $1.8\times$ improvement.

Table 15: **I/O Throughput and Latency Profiling on ImageNet-100.** We report representative per-iteration DataLoader profiling statistics. LGM relies on spatial view ensembling, whereas TPA uses K=1 and accumulates supervision temporally.

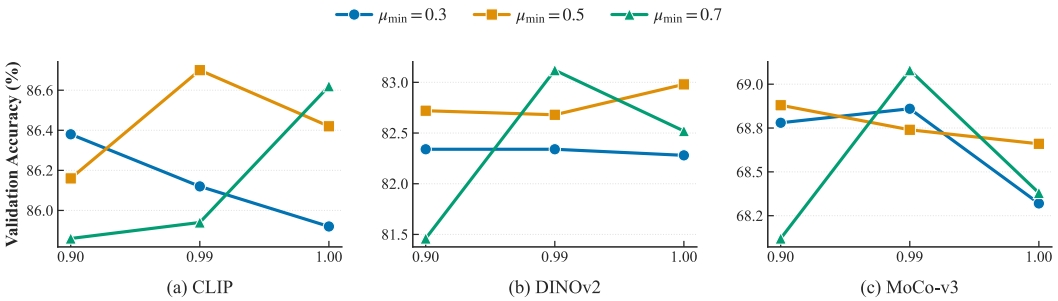
Method	Setup	I/O Load sec	Aug. Wait sec	Total Prep sec/iter	Throughput GB/s
LGM	CLIP	0.062	0.052	0.115	11.9
LGM	C+D+M	0.062	0.158	0.220	11.9
TPA	C+D+M	**0.008**	**0.034**	**0.043**	**21.6**

These profiling results show that the efficiency gain of TPA is not limited to lower GPU memory usage. By avoiding repeated spatial views within each iteration, TPA reduces input-pipeline waiting time and nearly doubles effective throughput. This supports the design choice of shifting redundancy reduction from simultaneous spatial averaging to temporal prototype accumulation, which improves scalability at both the compute and data-loading levels.

D.7 Relation to Train-from-Scratch Dataset Distillation

Our study focuses on distilling data for frozen pretrained feature extractors, where the synthetic set is used to train a linear classifier or to form nearest-prototype classifiers on top of fixed representations. This setting differs from traditional train-from-scratch dataset distillation, which optimizes synthetic images so that a randomly initialized neural network can be trained end-to-end from them. The two paradigms therefore impose different optimization targets and evaluation protocols: frozen-feature distillation measures how well synthetic images preserve class-discriminative semantics in pretrained representation spaces, whereas train-from-scratch distillation must additionally support feature learning from extremely limited data.

This distinction becomes especially important at ImageNet scale under IPC=1. Existing train-from-scratch methods can suffer severe optimization collapse in this regime. For example, RDED-style large-scale distilled data reports around 6.6% top-1 accuracy under the train-from-scratch evaluation protocol [29]. We therefore keep the main comparisons within the frozen-feature distillation setting, where LGM and our method share the same evaluation objective.



(a) CLIP (b) DINOv2 (c) MoCo-v3

Figure 7: **Hyperparameter Sensitivity Analysis of Temporal Gating.** We evaluate the linear probing accuracy on ImageNet-100 by varying the maximal gate threshold $\mu_{\max}$ and the minimum gate threshold $\mu_{\min}$ (lines). The results are reported on the source teacher (a) CLIP, and unseen target architectures (b) DINOv2 and (c) MoCo-v3. The default setting ($\mu_{\max} = 0.99$ and $\mu_{\min} = 0.5$) achieves the best trade-off between source-model performance and cross-model generalization.

D.8 Parameter Sensitivity

To verify the robustness of the temporal gating mechanism, we analyze the impact of the gate thresholds $\mu_{\max}$ and $\mu_{\min}$, as illustrated in Figure 7. The results show that the method is stable across a broad range of gate settings, with CLIP accuracy varying only moderately as the thresholds change. The best CLIP performance is obtained at $\mu_{\max} = 0.99$ and $\mu_{\min} = 0.5$, reaching 86.7%. This suggests that a high upper retention bound is useful for suppressing noisy single-view updates, while a moderate lower bound still allows the temporal prototype to adapt to reliable semantic signals. We therefore use $\mu_{\max} = 0.99$ and $\mu_{\min} = 0.5$ as the default configuration in our experiments.

Table 16: **Performance Analysis at IPC=5.** We report linear probing accuracy (%) with 5 images per class. Our method consistently outperforms the state-of-the-art LGM across all datasets. The significant improvement on the fine-grained Food-101 dataset demonstrates our method's ability to capture complex intra-class distributions.

Train Set (5 Img/Cls)	ImageNet-100				Food-101			
	CLIP	DINOv2	MoCo-v3	Average	CLIP	DINOv2	MoCo-v3	Average
Full Dataset	92.5±0.0	95.2±0.1	89.4±0.3	92.4±0.1	91.9±0.0	92.7±0.0	78.3±0.0	87.6±0.0
Neighbors	81.5±0.3	86.4±0.2	79.9±0.1	82.6±0.2	77.5±0.1	81.4±0.1	38.9±0.0	65.9±0.1
Centroids	82.5±0.2	87.6±0.2	79.0±0.1	83.0±0.4	78.0±0.4	80.5±0.3	32.0±0.3	63.5±0.0
Random	78.3±0.9	83.4±0.8	70.4±0.4	77.4±0.7	67.8±0.7	73.4±0.7	29.4±0.4	56.9±0.8
LGM	85.8±0.1	91.7±0.2	83.8±0.4	87.1±0.3	83.3±0.2	83.8±0.4	46.4±0.1	71.2±0.3
Ours$_{[C/D/M]}$	86.3±0.2	91.9±0.1	84.1±0.2	87.4±0.2	84.7±0.1	83.9±0.1	50.7±0.7	73.1±0.3
Ours$_{[C+D+M]}$	**87.9**±0.2	**92.6**±0.0	**86.7**±0.1	**89.1**±0.1	**85.3**±0.1	**84.7**±0.1	**54.4**±0.4	**74.8**±0.2

D.9 Quantitative and Qualitative Analysis at IPC=5

Table 16 provides the encoder-level IPC=5 results underlying Table 8, including full-data and sampling baselines. LGM uses $K = 2$ because of memory constraints on four RTX 4090 GPUs. TPA (C+D+M) improves over single-teacher TPA on every listed evaluation encoder for both datasets, indicating that the aggregate gains are shared across the three representations. Increasing IPC from 1 to 5 raises TPA (C+D+M) accuracy from 71.9% to 74.8% on Food-101 and from 88.8% to 89.1% on ImageNet-100. Thus, the observed capacity gain is larger on the fine-grained dataset, where multiple images per class can represent variations in appearance. On Food-101, the 2.9-point gain from increasing IPC and the 3.6-point advantage over LGM at IPC=5 describe complementary effects of synthetic capacity and distillation method. Table 9 further examines capacity under a fixed single-teacher protocol.

The synthesized images illustrate how the larger set represents variation within each class. Figure 13 shows structural and textural differences among Food-101 images, including variations in ingredients and presentation. On ImageNet-100, Figure 14 shows differences in object pose, background, and viewpoint among the five images of each class. These visualizations complement the encoder-level evaluation by showing the appearance variation represented in the synthetic sets.

E Additional Visualizations

We provide additional visualizations of the synthesized images. In Figure 8 and Figure 9, we present samples from Food-101 and ImageNet-100 distilled using the single CLIP teacher. Furthermore, to demonstrate the effectiveness of our method under the multi-teacher setting, we showcase results distilled using the ensemble of CLIP, DINOv2, and MoCo-v3. We illustrate these outcomes for Food-101 in Figure 10, for ImageNet-100 in Figure 11, and for Flowers-102 in Figure 12.

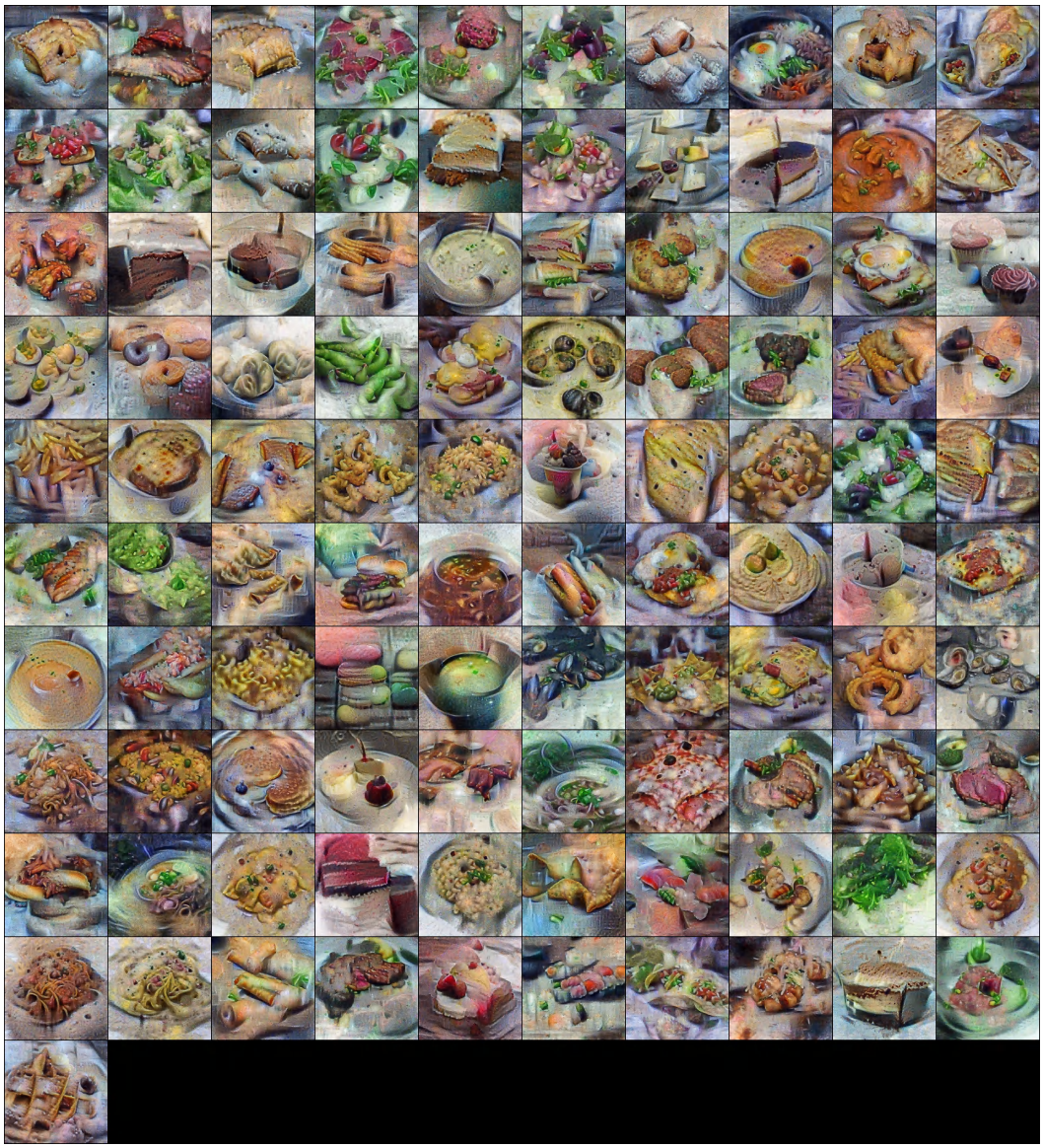


Figure 8: Food-101 distilled using CLIP.



Figure 9: ImageNet-100 distilled using CLIP.

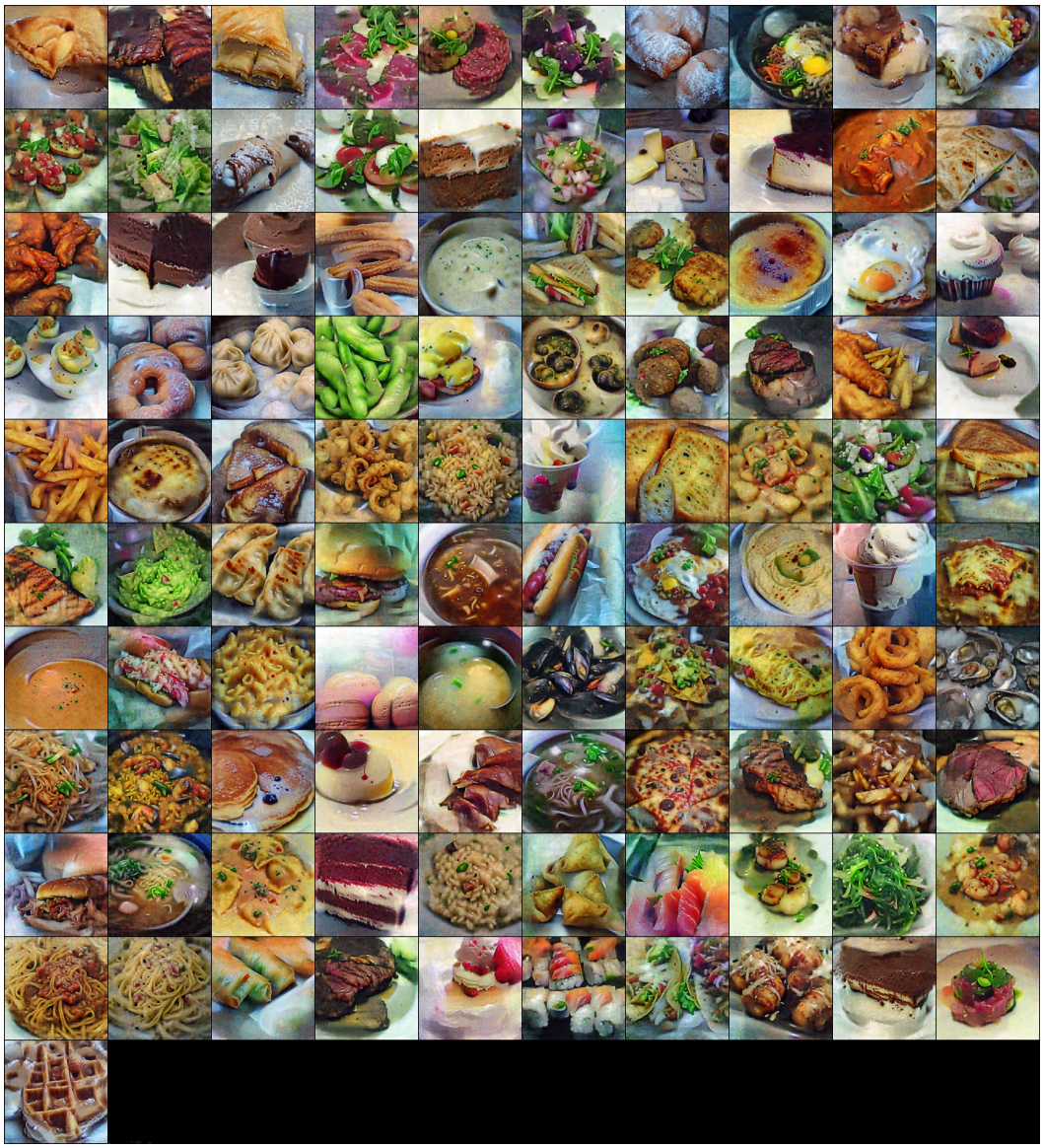


Figure 10: Food-101 distilled using CLIP, DINOv2, and MoCo-v3.

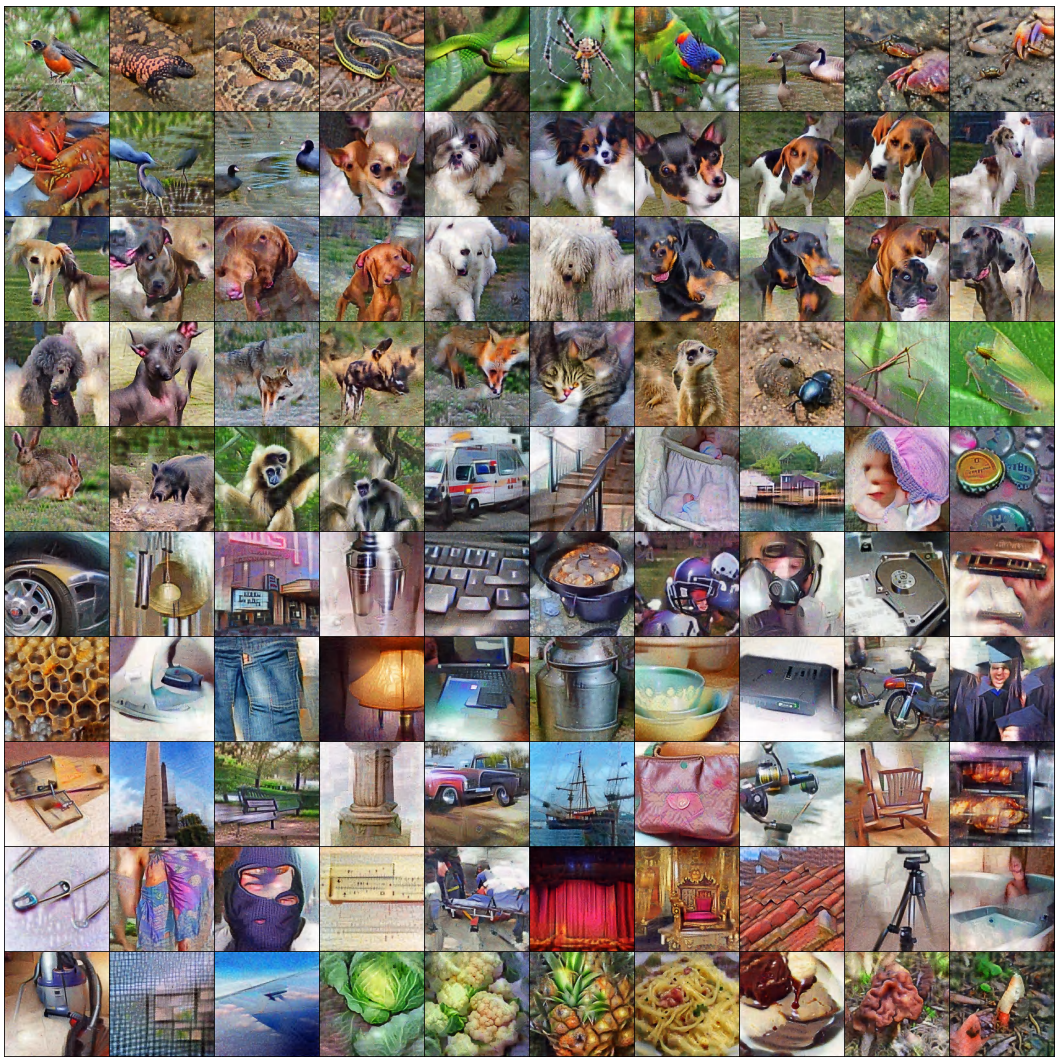


Figure 11: ImageNet-100 distilled using CLIP, DINOv2, and MoCo-v3.

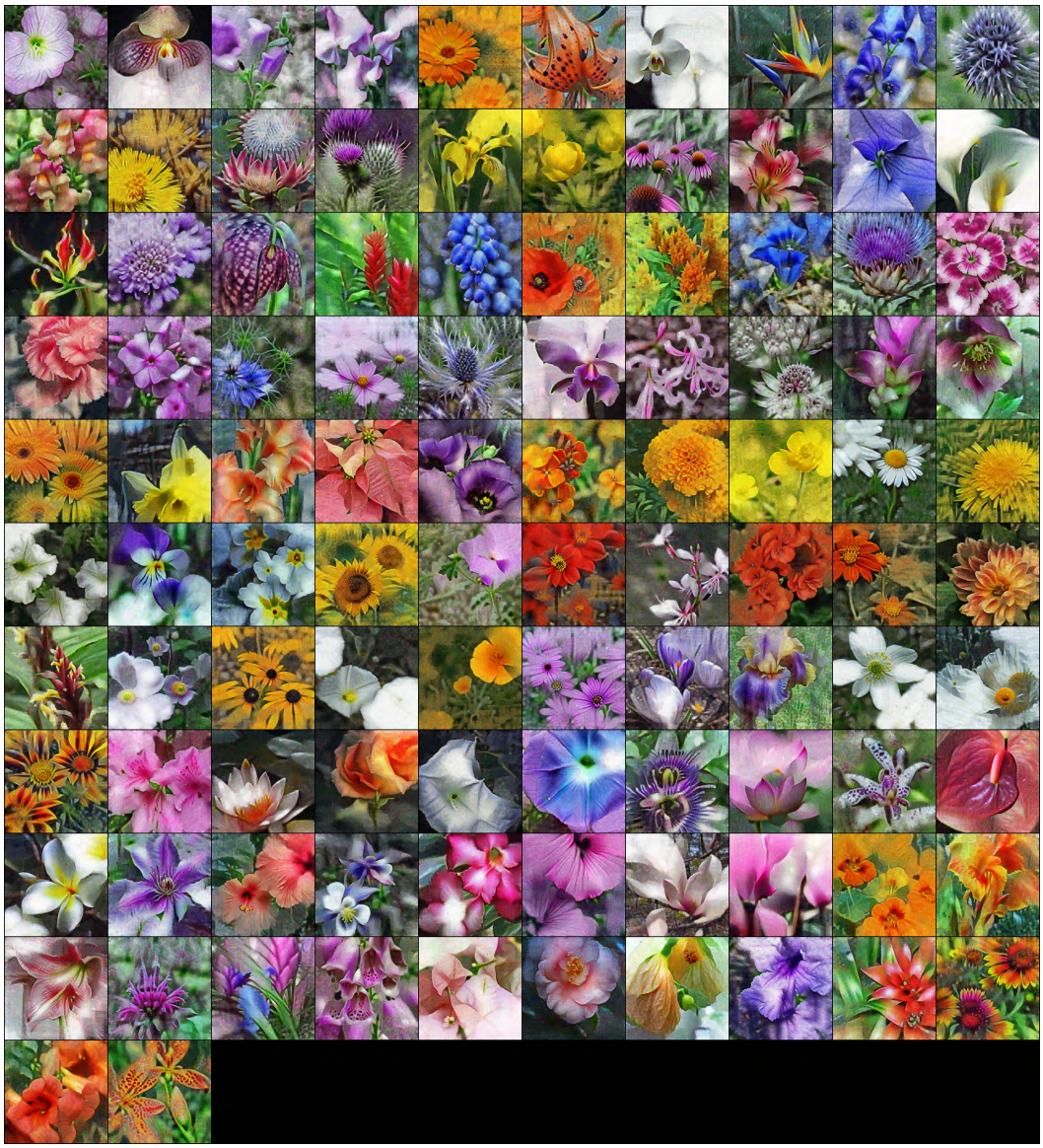


Figure 12: Flowers-102 distilled using CLIP, DINOv2, and MoCo-v3.

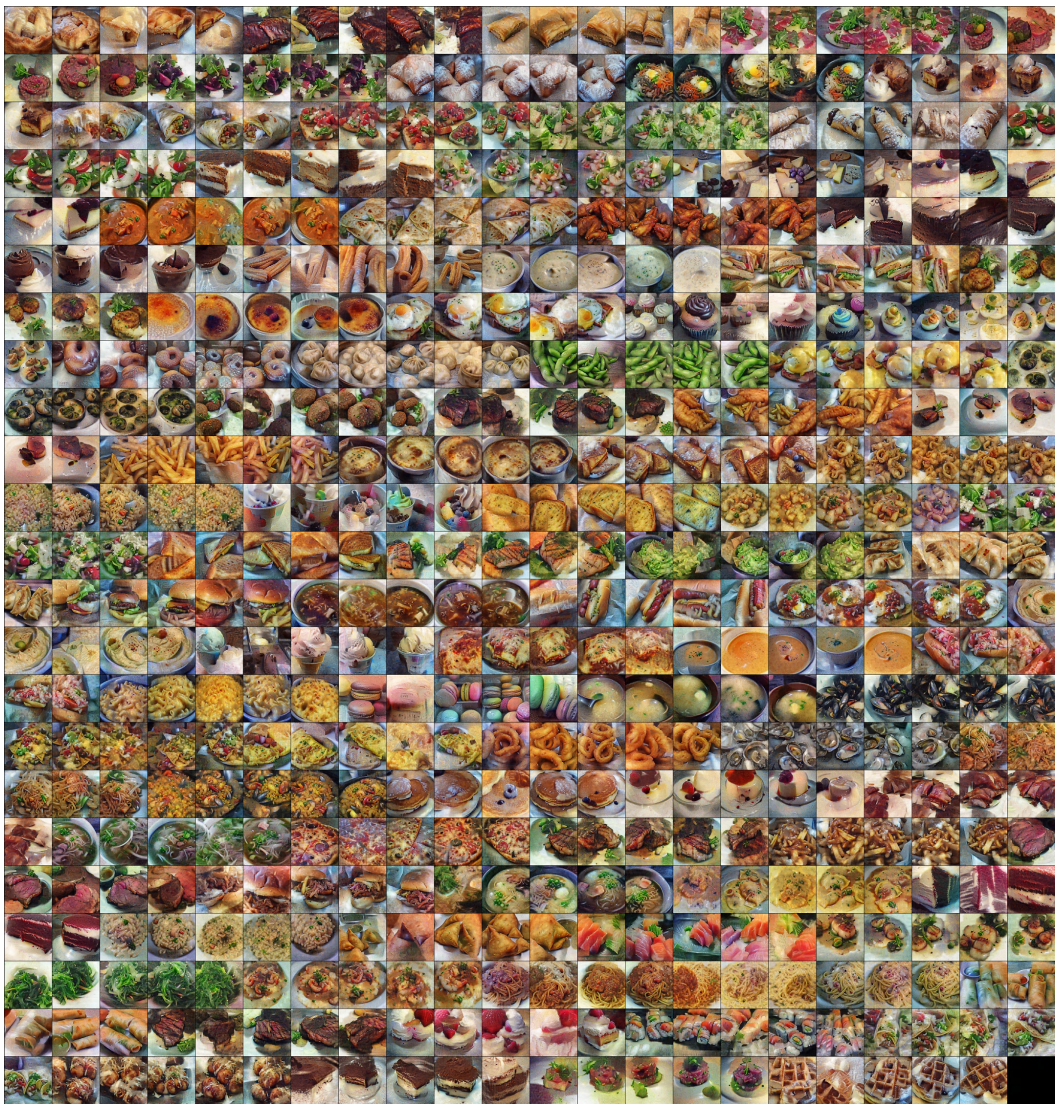


Figure 13: Food-101 distilled using CLIP, DINOv2, and MoCo-v3 (IPC=5).



Figure 14: ImageNet-100 distilled using CLIP, DINOv2, and MoCo-v3 (IPC=5).

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